

SUSY positivity research program

Mathematical background

Prepared 11 September 2026; expanded 13 September 2026.

This standalone background includes the definitions and derivations needed for its positive kinetic construction and its conditional routes to full Weil positivity. The full arithmetic matching remains an open construction problem.

1 Weil positivity, support length, and the shifted family

This document develops the common mathematical framework of the SUSY positivity research program. It gives the arithmetic target, the shift-energy identities and their domains, and an explicit positive auxiliary-field realization of the gamma kinetic term. Definitions and derivations used within this framework are included here. External references supply the classical Weil criterion, standard special-function identities, and the general theory of positive extensions. A positive realization of the full arithmetic form at arbitrary support length remains the objective, not a conclusion of the constructions below.

The positivity problem has two parameters with different roles. The length L specifies the support available to a test function. The shift ω belongs to an auxiliary family of zeta transfers and their generators. The classical equivalence with the Riemann hypothesis (RH) concerns the central form, at $\omega = 0$, on inputs of arbitrary compact support. We first state that target, then explain why shifted forms and contraction identities are useful in seeking a positive realization of it. We also distinguish a direct amplitude factor, a bulk energy reduced to boundary data, and an energy accumulated along the shift family.

1.1 The arithmetic form and its input space

Let $I_L = (-L/2, L/2)$, so that $L > 0$ is the total interval length. All norms and inner products below use unweighted Lebesgue measure, with inner products conjugate-linear in the first variable. Extend $f \in L^2(I_L)$ by zero to $\tilde{f}$ on $\mathbb{R}$, and set

$$\hat{f}(\tau) = \int_{I_L} f(x) e^{-i\tau x} dx, \quad \mathcal{D}_{\log, L} = \left\{ f \in L^2(I_L) : \int_{\mathbb{R}} \log(2 + |\tau|) |\hat{f}(\tau)|^2 d\tau < \infty \right\}. \quad (1.1)$$

For $a > 0$ write $(T_a f)(x) = \tilde{f}(x - a)$ on I_L , and define

$$C(f) = \int_{I_L} f(x) \cosh(x/2) dx, \quad S(f) = \int_{I_L} f(x) \sinh(x/2) dx.$$

The central finite-interval Weil form in our normalization is

$$\begin{aligned} Q_{0,L}[f] = & \frac{1}{2\pi} \int_{\mathbb{R}} \left(\operatorname{Re} \psi \left(\frac{1}{4} + \frac{i\tau}{2} \right) - \log \pi \right) |\hat{f}(\tau)|^2 d\tau \\ & + 2|C(f)|^2 - 2|S(f)|^2 - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (1.2)$$

Here $\psi = \Gamma'/\Gamma$ and Λ is the von Mangoldt function. The sum is finite and only prime powers contribute. A delay of length L has zero action, which explains the strict cutoff.

The multiplier in (1.2) differs from $\log(2+|\tau|)$ by a bounded function, by the digamma asymptotics [2, §5.11]. The remaining terms are bounded forms at each fixed L . Consequently $Q_{0,L}$ is closed and semibounded on $\mathcal{D}_{\log,L}$ without any positivity assumption. Smooth compactly supported inputs form a core: dilation can first move the support into the interval interior, and mollification then approximates in the weighted Fourier norm in (1.1).

We write $Q_{0,L}^\gamma$ for the form obtained by omitting the explicit prime sum. This convention retains the normalization and pole terms $2|C|^2 - 2|S|^2$. It gives $Q_{0,L}^\gamma = Q_{0,L}$ for $L \leq \log 2$; beyond that range gamma positivity alone does not establish the required arithmetic positivity.

Separating the positive kinetic term from the full form. The gamma multiplier has a frequency-dependent part and a constant part. Define, for $s \geq 0$,

$$B(s) = \operatorname{Re} \psi\left(\frac{1}{4} + \frac{i\sqrt{s}}{2}\right) - \psi\left(\frac{1}{4}\right), \quad w_0 = \psi\left(\frac{1}{4}\right) - \log \pi.$$

The constant is negative: $w_0 = -\gamma_E - \pi/2 - 3\log 2 - \log \pi < 0$, where γ_E is Euler's constant. For the zero-extended input, set

$$K[\tilde{f}] = \frac{1}{2\pi} \int_{\mathbb{R}} B(\tau^2) |\hat{f}(\tau)|^2 d\tau, \quad P[f] = 2|C(f)|^2 - 2|S(f)|^2.$$

The letter B denotes a scalar frequency response; K denotes the quadratic functional obtained by weighting all Fourier components with that response. The argument $s = \tau^2$ is squared spatial frequency, not a time or a zeta shift.

Write $\ell_p = \log p$ and $r_p = p^{-1/2}$ for a prime p . Since $\Lambda(p^m) = \log p$ and $\Lambda(n) = 0$ off prime powers, Plancherel's identity rewrites the full central form exactly as

$$\boxed{\begin{aligned} Q_{0,L}[f] &= K[\tilde{f}] + w_0 \|f\|^2 + P[f] \\ &\quad - \sum_{\substack{p \text{ prime}, m \geq 1 \\ m\ell_p < L}} \ell_p r_p^m \langle f, (T_{m\ell_p} + T_{m\ell_p}^*) f \rangle. \end{aligned}}$$

In particular,

$$Q_{0,L}^\gamma[f] = K[\tilde{f}] + w_0 \|f\|^2 + P[f].$$

Thus the gamma kinetic energy is K , while the gamma form in this document includes the contact and pole terms as well. Even on a prime-free interval, the full form is generally not equal to K . This decomposition is an arithmetic identity; it does not assert positivity of its signed terms or of their sum.

The positive structure of B follows from the digamma partial-fraction identity [2, §5.7]. With $a_k = 2k + 1/2$,

$$\begin{aligned} B(\tau^2) &= \sum_{k=0}^{\infty} \left(\frac{1}{k + 1/4} - \frac{k + 1/4}{(k + 1/4)^2 + \tau^2/4} \right) \\ &= \sum_{k=0}^{\infty} \frac{2}{a_k} \frac{\tau^2}{a_k^2 + \tau^2} \geq 0. \end{aligned}$$

Each difference is nonnegative, and the series converges with terms $O_\tau(k^{-3})$. Therefore $K \geq 0$ on its form domain. The later bulk-boundary construction derives each summand as the minimum of a positive field energy. The other terms in the boxed decomposition specify what a full positive realization must still reproduce.

1.2 The quantifiers in the RH criterion

Weil's positivity criterion, in its smooth compact-support formulation [1, §1.1], states

$$\boxed{\text{RH} \iff Q_{0,L}[f] \geq 0 \text{ for every } L > 0 \text{ and every } f \in C_c^\infty(I_L).} \quad (1.3)$$

Complex-valued test functions are allowed. Every compactly supported smooth function on $\mathbb{R}$ belongs to $C_c^\infty(I_L)$ for some L , and (1.2) agrees on that input at every larger length. Conversely, every such finite-interval input is a whole-line test function. This explains the equivalence between the usual global criterion and (1.3). By the core property, one may equivalently require nonnegativity on every $\mathcal{D}_{\log,L}$.

In particular, the first subscript in $Q_{0,L}$ has already fixed $\omega = 0$. The remaining quantifiers range over the length and *all* inputs, including inputs outside any chosen finite-dimensional approximation. Positivity here means nonnegativity. A lower bound $Q_{0,L}[f] \geq m_L \|f\|^2$ with $m_L > 0$ is useful for quantitative arguments, but the criterion requires no positive constant independent of L .

An unbounded sequence of covered lengths suffices: any fixed input eventually fits in one of them. A finite collection of lengths does not suffice, and decomposing a long input into short pieces introduces cross terms that must also be controlled.

1.3 The shifted transfer and its generator

Define the completed zeta function and its central translate by

$$\xi(s) = \frac{1}{2} s(s-1) \pi^{-s/2} \Gamma(s/2) \zeta(s), \quad F(p) = \xi(\tfrac{1}{2} + p).$$

For $0 < \omega \leq 1/2$ the causal transfer has Laplace symbol

$$K_\omega(p) = \frac{F(p-\omega)}{F(p+\omega)}, \quad \Theta_\omega(z) = K_\omega(-iz) = \frac{\xi(1/2 - \omega - iz)}{\xi(1/2 + \omega - iz)}. \quad (1.4)$$

The causal realization is defined on a line $\text{Re } p = \eta > 1$, where the relevant zeta Dirichlet series converge absolutely and the denominator is nonzero. Let k_ω be its inverse Laplace kernel, supported in $[0, \infty)$. Then $V_{\omega,L} f = (k_\omega * \tilde{f})|_{I_L}$. Equivalently, identify interval inputs with their zero extensions, let P_L multiply by $\mathbf{1}_{I_L}$ on the whole line, and let M_η multiply by $e^{-\eta x}$. Then

$$V_{\omega,L} = M_\eta^{-1} P_L \mathcal{M}[K_\omega(\eta + i \cdot)] P_L M_\eta,$$

where $\mathcal{M}[b]$ denotes the whole-line Fourier multiplier by b . This defines a bounded finite-interval operator; it does not assume a bounded whole-line transfer on the unweighted space. At zero shift $K_0 = 1$ and $V_{0,L} = I$ as a strong limit.

The symmetric generator form can be specified directly on $\mathcal{D}_{\log,L}$. Put $R(t) = 2 \cosh(t/2) - e^{-t/2}/(1 - e^{-2t})$ for $t > 0$. Then

$$\begin{aligned} Q_{\omega,L}[f] &= Q_{0,L}[f] + \int_{I_L^2} (\cosh(\omega|x-y|) - 1) R(|x-y|) \overline{f(x)} f(y) dx dy \\ &\quad - \sum_{\substack{n \geq 2 \\ \log n < L}} \frac{\Lambda(n)}{\sqrt{n}} (\cosh(\omega \log n) - 1) \langle f, (T_{\log n} + T_{\log n}^*) f \rangle. \end{aligned} \quad (1.5)$$

The added integral kernel is $O(|x-y|)$ at the diagonal. Together with the finite delay sum this yields, for each fixed L , a constant C_L such that

$$|Q_{\omega,L}[f] - Q_{0,L}[f]| \leq C_L \omega^2 \|f\|^2, \quad 0 \leq \omega \leq \tfrac{1}{2}. \quad (1.6)$$

The bound concerns the bounded difference of two generally unbounded forms. Formula (1.5) also specifies the endpoint $\omega = 1/2$ without dropping a distributional correction.

The generator, explicitly. Logarithmic differentiation of (1.4) gives the causal generator symbol

$$a_\omega(p) = \frac{F'(p - \omega)}{F(p - \omega)} + \frac{F'(p + \omega)}{F(p + \omega)}, \quad \partial_\omega K_\omega = -a_\omega K_\omega.$$

The generator $G_{\omega,L}$ is realized from a_ω exactly as $V_{\omega,L}$ is realized from K_ω , on the same line $\operatorname{Re} p = \eta > 1$ and with the same compression:

$$G_{\omega,L} = M_\eta^{-1} P_L \mathcal{M}[a_\omega(\eta + i \cdot)] P_L M_\eta,$$

with P_L multiplication by $\mathbf{1}_{I_L}$, M_η multiplication by $e^{-\eta x}$, and $\mathcal{M}[b]$ the whole-line Fourier multiplier by b . Equivalently, $G_{\omega,L}g = (g_\omega * \tilde{g})|_{I_L}$ for the inverse Laplace kernel g_ω of a_ω , supported in $[0, \infty)$. The natural domain is $\mathcal{D}_{\log,L}$ of (1.1), on which $Q_{\omega,L}$ is closed and semibounded; $G_{\omega,L}$ is not symmetric, and only its symmetric part is a form:

$$\operatorname{Re}\langle g, G_{\omega,L}g \rangle = Q_{\omega,L}[g], \quad \text{equivalently} \quad G_{\omega,L}^* + G_{\omega,L} \text{ represents } 2Q_{\omega,L}.$$

This is the *complete* shifted form, not a part of it. Writing ξ as its three factors, with $s = 1/2 + p$,

$$\frac{F'}{F}(p) = \underbrace{\frac{1}{p + 1/2} + \frac{1}{p - 1/2}}_{\frac{1}{2}s(s-1)} \underbrace{-\frac{1}{2}\log\pi + \frac{1}{2}\psi\left(\frac{1}{4} + \frac{p}{2}\right)}_{\pi^{-s/2}\Gamma(s/2)} + \underbrace{\frac{\zeta'}{\zeta}\left(\frac{1}{2} + p\right)}_{\zeta(s)},$$

and each factor supplies one term of (1.2) and (1.5).

The zeta factor. Absolute convergence on $\operatorname{Re} p = \eta > 1$ gives $\frac{\zeta'}{\zeta}(\frac{1}{2} + p - \omega) + \frac{\zeta'}{\zeta}(\frac{1}{2} + p + \omega) = -2 \sum_{n \geq 2} \Lambda(n) n^{-1/2} \cosh(\omega \log n) n^{-p}$, and n^{-p} is the causal delay by $\log n$. So the zeta factor contributes $-2 \sum_{\log n < L} \Lambda(n) n^{-1/2} \cosh(\omega \log n) T_{\log n}$ before symmetrization, the cutoff being automatic because a delay of length L has zero action on I_L .

The gamma and π factors. With $\psi(z) = -\gamma_E + \int_0^\infty (e^{-t} - e^{-zt})(1 - e^{-t})^{-1} dt$ at $z = \frac{1}{4} + \frac{p \mp \omega}{2}$ and $t = 2r$, these contribute a constant together with the causal kernel $-2 \cosh(\omega r) e^{-r/2} (1 - e^{-2r})^{-1}$ at separation r . At $\omega = 0$ and $p = i\tau$ the multiplier is $\operatorname{Re} \psi(\frac{1}{4} + \frac{i\tau}{2}) - \log \pi$, the first term of (1.2).

The pole factors. The four simple poles of a_ω at $p = \pm \frac{1}{2} \pm \omega$ have causal kernel $4 \cosh(\omega x) \cosh(x/2) \mathbf{1}_{x>0}$, which under $\operatorname{Re}\langle g, \cdot g \rangle$ becomes $2 \cosh(\omega(x - y)) \cosh(\frac{x-y}{2})$; at $\omega = 0$ this is $2|C(g)|^2 - 2|S(g)|^2$, the contact and pole term of (1.2).

Collecting the three, the off-diagonal kernel of $Q_{\omega,L}$ away from the prime atoms is $\cosh(\omega(x - y))R(|x - y|)$ with R as in (1.5), and each prime atom carries $-\Lambda(n)n^{-1/2} \cosh(\omega \log n)$. Both are the $\omega = 0$ kernels multiplied by $\cosh(\omega \cdot)$, which is (1.10) seen on the kernel side, and both leave the diagonal unchanged since $\cosh 0 = 1$.

The pole term is invisible on the imaginary axis. The causal definition on $\operatorname{Re} p = \eta > 1$ is not a technical convenience here. At $p = i\tau$,

$$\operatorname{Re} \left[\frac{2}{p - 1/2} + \frac{2}{p + 1/2} \right] = \frac{2 \cdot \frac{1}{2}}{\frac{1}{4} + \tau^2} + \frac{2 \cdot (-\frac{1}{2})}{\frac{1}{4} + \tau^2} = 0,$$

identically in τ . Reading a_ω as a Fourier multiplier on the imaginary axis therefore recovers the archimedean term and the prime delays and *loses the contact and pole term entirely*. The pole factors reach $Q_{\omega,L}$ only through the right-half-plane Laplace realization above.

The gamma kinetic term as an infinitesimal shift response. Isolating the gamma and π factors of K_ω gives

$$K_\omega^\Gamma(p) = \pi^\omega \frac{\Gamma(1/4 + p/2 - \omega/2)}{\Gamma(1/4 + p/2 + \omega/2)}.$$

Direct logarithmic differentiation yields

$$-\partial_\omega \log K_\omega^\Gamma(p) \Big|_{\omega=0} = \psi(1/4 + p/2) - \log \pi.$$

The gamma expression is regular at $p = i\tau$, so its real part is

$$\operatorname{Re} \left[-\partial_\omega \log K_\omega^\Gamma(i\tau) \Big|_{\omega=0} \right] = B(\tau^2) + w_0.$$

This is the arithmetic origin of the kinetic multiplier. The positive tower realizes the B portion of the zero-shift generator, not the finite-shift transfer itself. Its fields therefore remain nontrivial at $\omega = 0$, although $K_0 = 1$ and $D_{0,L} = 0$, with the contraction defect $D_{\omega,L}$ defined below. Evaluating this gamma expression on the imaginary axis does not replace the right-half-plane causal definition of the full zeta transfer.

For fixed L , first choose a test function $f \in C_c^\infty(I_L)$: a smooth, possibly complex-valued input compactly supported inside I_L . Keep f fixed as ω varies. The evolution and its scalar energy balance are

$$\begin{aligned} \frac{d}{d\omega} (V_{\omega,L} f) &= -G_{\omega,L} (V_{\omega,L} f), \\ \frac{d}{d\omega} \|V_{\omega,L} f\|_{L^2(I_L)}^2 &= -2Q_{\omega,L} [V_{\omega,L} f]. \end{aligned} \tag{1.7}$$

The norm in the second line is the usual unweighted $L^2(I_L)$ norm, so its square is $\int_{I_L} |(V_{\omega,L} f)(x)|^2 dx$. The brackets in $Q_{\omega,L} [V_{\omega,L} f]$ mean evaluation: first apply the linear operator $V_{\omega,L}$ to f , then evaluate the quadratic functional $Q_{\omega,L}$ on the resulting function to obtain a scalar energy. This is composition as maps, rather than a product of two linear operators. Products of operators below, such as $G_{\omega,L} V_{\omega,L}$, do denote operator composition.

The first identity holds for every such f , so we may express it as an *operator differential equation*:

$$\boxed{\frac{\partial V_{\omega,L}}{\partial \omega} = -G_{\omega,L} V_{\omega,L}, \quad V_{0,L} = I.}$$

Here f has become implicit: the operator identity means that the two sides have equal action on every admissible test function. It describes the evolution of the entire transformation $V_{\omega,L}$, with $V_{0,L} f = f$ as its initial condition.

The energy balance also has an operator version. The identity

$$\|V_{\omega,L} f\|^2 = \langle f, V_{\omega,L}^* V_{\omega,L} f \rangle$$

identifies $V_{\omega,L}^* V_{\omega,L}$ as the operator representing the output norm squared. At positive shift, norm differentiability as justified below permits the product rule:

$$\begin{aligned} \frac{\partial}{\partial \omega} (V_{\omega,L}^* V_{\omega,L}) &= \left(\frac{\partial V_{\omega,L}}{\partial \omega} \right)^* V_{\omega,L} + V_{\omega,L}^* \frac{\partial V_{\omega,L}}{\partial \omega} \\ &= -(G_{\omega,L} V_{\omega,L})^* V_{\omega,L} - V_{\omega,L}^* (G_{\omega,L} V_{\omega,L}). \end{aligned}$$

Consequently the contraction defect $D_{\omega,L} = I - V_{\omega,L}^* V_{\omega,L}$ satisfies the operator energy balance

$$\boxed{\frac{\partial D_{\omega,L}}{\partial \omega} = V_{\omega,L}^* (G_{\omega,L}^* + G_{\omega,L}) V_{\omega,L}, \quad D_{0,L} = 0.}$$

In this last formula the right-hand side is understood through quadratic forms: $G_{\omega,L}^* + G_{\omega,L}$ represents the symmetric generator form $2Q_{\omega,L}$, pulled back by $V_{\omega,L}$. The preceding product-rule expression specifies the bounded operator at positive shift without requiring the separate application of $G_{\omega,L}^*$ to an evolved input. Pairing with any admissible f recovers the scalar energy balance in (1.7); conversely, its validity for every test function determines the symmetric form by polarization. Thus f is implicit in the operator energy balance as well.

At $\omega = 0$ the evolution identity is interpreted by a right derivative on these test functions, and $V_{0,L} = I$ is a strong limit. No operator-norm derivative at zero is asserted. To justify these statements, the gamma ratio asymptotics and absolutely convergent Dirichlet series on $p = \eta + i\tau$ give, uniformly for $0 \leq \omega \leq 1/2$,

$$|\partial_\omega^j K_\omega(\eta + i\tau)| \leq C_\eta(1 + \log(2 + |\tau|))^j(2 + |\tau|)^{-\omega}, \quad j = 0, 1, 2.$$

For positive shift these bounds give differentiability and smoothing into H^r for $0 < r < \min(\omega, 1/2)$, hence into the form domain. Causality permits compression of the generator identity to I_L . For smooth compactly supported initial data, rapid Fourier decay permits differentiation at zero by dominated convergence. Integrating from a positive shift and then taking its limit to zero gives the identities below. No bounded inverse of $V_{\omega,L}$ is needed.

1.4 Instantaneous energy, cumulative contraction, and the zero-shift limit

For the contraction defect just defined, integrating the energy balance in (1.7) on smooth compactly supported inputs yields

$$\langle f, D_{\omega,L}f \rangle = \|f\|^2 - \|V_{\omega,L}f\|^2 = 2 \int_0^\omega Q_{s,L}[V_{s,L}f] ds, \quad (1.8)$$

$$Q_{0,L}[f] = \lim_{\omega \downarrow 0} \frac{\langle f, D_{\omega,L}f \rangle}{2\omega}. \quad (1.9)$$

Here s is a running shift. The generator controls instantaneous change, whereas $D_{\omega,L}$ records the accumulated change along the moving input $V_{s,L}f$. Nonnegativity of $D_{\omega,L}$ on every input is exactly the contraction condition $\|V_{\omega,L}\| \leq 1$. Although $D_{0,L} = 0$, its initial derivative contains the nontrivial central form.

Positivity of $Q_{\omega,L}$ at all shifts is not the criterion (1.3). It is an additional requirement on the family. Indeed, with $Xf(x) = xf(x)$, expansion of the kernels gives

$$Q_{\omega,L}[f] = Q_{0,L}[\cosh(\omega X)f] - Q_{0,L}[\sinh(\omega X)f]. \quad (1.10)$$

This is a difference of central energies; their individual nonnegativity does not supply the comparison needed for a nonnegative difference. Similarly, a negative instantaneous value on one input away from zero does not determine the sign of the cumulative integral (1.8).

A useful sufficient local route starts with $Q_{0,L} \geq m_L I$, where $m_L > 0$. Equations (1.6) and (1.7) imply $Q_{s,L} \geq (m_L - C_L s^2)I$, and integration gives

$$\|V_{\omega,L}\| \leq \exp(-m_L \omega + C_L \omega^3/3). \quad (1.11)$$

Thus sufficiently small positive shifts give contraction. Their allowed range may shrink with L ; no common positive shift range is required for the following implication.

The central-limit identity and restriction of causal outputs to shorter intervals give the following sufficient route directly.

Proposition 1.1 (A sufficient contraction route to RH). *If there exist $L_j \rightarrow \infty$ and $0 < \omega_j \downarrow 0$ with $\|V_{\omega_j, L_j}\| \leq 1$ for every j , then RH holds.*

Proof. Fix $L > 0$ and $f \in C_c^\infty(I_L)$. For sufficiently large j , extend f by zero into I_{L_j} and restrict its output back to I_L . The causal convolution defining the transfers gives $\|V_{\omega_j, L} f\| \leq \|f\|$. The quotients in (1.9) are therefore nonnegative along this sequence. Their limit gives $Q_{0,L}[f] \geq 0$. The input and length were arbitrary, so (1.3) applies. $\square$

The shift disappears from the conclusion because it has been sent to zero. This proposition is an auxiliary sufficient route; the central criterion itself is the equivalence (1.3). If central positivity has already been established at arbitrarily large lengths, the contraction argument is unnecessary.

1.5 Positive factors as one route within the program

The direct RH objective of a positive realization is an exact identity

$$Q_{0,L}[f] = \|A_{0,L}f\|_{\mathcal{K}_{0,L}}^2, \quad f \in C_c^\infty(I_L), \quad (1.12)$$

with $\mathcal{K}_{0,L}$ a positive Hilbert space and $A_{0,L}$ defined from arithmetic data without assuming the sign of the target form. The zero in both subscripts fixes $\omega = 0$: this is the central identity, not a claim that the construction cannot depend on a shift. The shorter notation A_L would mean $A_{0,L}$ here. If such a construction and its matching identity hold at arbitrary lengths, (1.3) proves RH directly. Separate finite-interval factors suffice for this implication; a single closable operator on ordinary whole-line L^2 is an additional operator-realization requirement.

A shifted factorization would instead have the form

$$Q_{\omega,L}[f] = \|A_{\omega,L}f\|_{\mathcal{K}_{\omega,L}}^2, \quad (\omega, L) \in \mathcal{R}, \quad (1.13)$$

where $\mathcal{R}$ is the region of parameters on which this identity is proved, and the inputs range over the stated form domain. Its restriction at $\omega = 0$ is (1.12). We do not require $\mathcal{R}$ to contain every shift at every length. For the direct RH route, it is enough to cover $(0, L)$ at arbitrarily large lengths. A model defined only for positive shifts would need a proved central limit, or the contraction route in Proposition 1.1.

Supersymmetric notation organizes a suitable closed factor into an off-diagonal supercharge and a nonnegative partner Hamiltonian. The arithmetic content lies in deriving (1.12). Choosing the target's spectral square root after assuming its positivity does not provide that derivation.

A full factor proved only on a bounded range of support lengths has that restricted scope. A factor of K alone, or an identity on a proper subspace of inputs, also requires an additional argument to reach the full central criterion. The gamma kinetic factor is derived explicitly below. A full shifted factor, where available, would enter the accumulated-energy identity at the end of the bulk discussion; no such factor is assumed in deriving the shift-energy balance itself.

1.6 What a bulk–boundary realization would mean

The factor $A_{\omega,L}$ need not act through a local expression in the input coordinate alone. It could arise by solving for auxiliary fields and evaluating their positive energy. In this interpretation, the *boundary data* are the prescribed input f and the *bulk variables* are the additional degrees of freedom allowed to adjust to that input. The construction has three steps: prescribe the boundary input, solve for the bulk configuration of least energy, and express that minimum using the boundary input alone. The last step can produce a nonlocal quadratic form even when the unreduced energy involves only local fields and derivatives.

What the boundary map does. Let $\mathcal{X}_{\omega,L}$ be a linear space of admissible bulk configurations, with its specified regularity and finite-energy conditions. A linear boundary map

$$\mathrm{Tr} : \mathcal{X}_{\omega,L} \longrightarrow \mathcal{B}_L, \quad U \longmapsto \mathrm{Tr} U$$

extracts the input visible at the boundary. Here $\mathcal{B}_L$ is the space of allowed boundary functions, with a topology appropriate to the construction; it must contain the test functions required by the proposed matching identity. The notation denotes a *trace map*, not the matrix or operator trace.

For a geometric field, the model case is $(\mathrm{Tr} U)(x) = U(x, 0)$. The variable x still runs over the input interval: the boundary datum is the entire function $f(x)$, not just its values at $x = \pm L/2$. If the bulk occupies $\mathbb{R} \times (0, \infty)$, one can prescribe the whole lower-edge trace to be the zero extension $\tilde{f}$; we then identify that trace with f on I_L . For nonsmooth fields, boundary restriction must be defined as a continuous trace in the chosen energy-space topology, rather than by pointwise evaluation.

For an auxiliary-channel configuration $U = (f, u_0, u_1, \dots)$, the same notation simply means the projection $\mathrm{Tr} U = f$. There need not be an additional continuous coordinate or a geometric edge. In either case, the trace selects the prescribed data; it does not determine the other components of U . Their determination is the role of the energy minimization. Conditions at other edges or at infinity belong to the bulk domain and must also be specified.

Fixing the boundary and minimizing the interior. Suppose the system has an independently defined nonnegative quadratic energy $\mathcal{E}_{\omega,L}[U]$. For each fixed f , consider *all* admissible configurations satisfying $\mathrm{Tr} U = f$. Only the bulk variables vary in this minimization; f , ω and L remain fixed. The proposed arithmetic matching is

$$Q_{\omega,L}[f] \stackrel{\text{target}}{=} \mathcal{E}_{\omega,L}^{\text{eff}}[f] := \inf_{\mathrm{Tr} U = f} \mathcal{E}_{\omega,L}[U]. \quad (1.14)$$

The effective energy is the least cost of sustaining that boundary profile. A chosen extension gives an upper bound on this cost; it gives the exact cost only when it minimizes the energy. The admissible class must be nonempty and contain a finite-energy configuration for every required f . Since every competing energy is nonnegative, its infimum is nonnegative as well. Existence of an actual minimizer is a further analytical question, not a consequence of nonnegativity alone.

To see how one finds the minimizer, temporarily suppress ω, L . Write $\mathcal{E}[U] = e(U, U)$, where e is a Hermitian sesquilinear form. Choose any admissible lift Rf with $\mathrm{Tr}(Rf) = f$. Every competitor then has the form

$$U = Rf + h, \quad h \in \mathcal{N} := \ker \mathrm{Tr}.$$

Thus $\mathcal{N}$ consists precisely of variations that leave the boundary fixed. If $\mathcal{P}f$ is a minimizer, variations by th and ith , with real t , give the weak equilibrium equations

$$\mathrm{Tr}(\mathcal{P}f) = f, \quad e(h, \mathcal{P}f) = 0 \quad \text{for every } h \in \mathcal{N}.$$

The first equation prescribes the boundary; the second says that the energy cannot be lowered by changing only the interior. It is the Euler–Lagrange equation of the bulk problem. In fact, for every $h \in \mathcal{N}$ it gives the exact decomposition

$$\mathcal{E}[\mathcal{P}f + h] = \mathcal{E}[\mathcal{P}f] + \mathcal{E}[h],$$

which verifies minimality directly.

One sufficient setting for a unique linear extension $f \mapsto \mathcal{P}f$ is a Hilbert bulk space with continuous trace, a bounded linear lift R from the chosen boundary space, and a continuous form e satisfying $e(h, h) \geq c\|h\|_{\mathcal{X}}^2$ on $\mathcal{N}$ for some $c > 0$. The correction h_f is then the unique solution of $e(h, h_f) = -e(h, Rf)$ for all $h \in \mathcal{N}$, and $\mathcal{P}f = Rf + h_f$ is linear in f . These are sufficient hypotheses, not assumptions already proved for an arithmetic completion. In degenerate or infinite-channel models, one must instead justify the relevant completion, possible zero-energy quotient, and attainment on the stated domain.

From the minimizing field to a boundary factor. Restore the parameters and suppose the bulk energy is given independently as $\mathcal{E}_{\omega,L}[U] = \|\mathcal{C}_{\omega,L}U\|_{\mathcal{K}_{\omega,L}}^2$. For a local field theory, $\mathcal{C}_{\omega,L}U$ can be a list of weighted derivatives and field combinations; for a network, it can be a list of edge differences and local amplitudes. If a linear minimizing extension $\mathcal{P}_{\omega,L}$ exists, the construction reads

$$f \xrightarrow{\mathcal{P}_{\omega,L}} U_{\min} \xrightarrow{\mathcal{C}_{\omega,L}} \text{energy amplitudes}, \quad \text{Tr } \mathcal{P}_{\omega,L} = I.$$

Consequently

$$A_{\omega,L} := \mathcal{C}_{\omega,L}\mathcal{P}_{\omega,L}, \quad \mathcal{E}_{\omega,L}^{\text{eff}}[f] = \|A_{\omega,L}f\|^2.$$

Here the trace goes from bulk to boundary, whereas the minimizing extension goes from boundary to bulk. Their composition above is the identity on boundary inputs; $\mathcal{P}_{\omega,L} \text{Tr}$ generally replaces a bulk field by its equilibrium extension rather than returning that field unchanged. The factor $A_{\omega,L}$ records the energy amplitudes of this extension. Its definition comes from solving the bulk equations, and can therefore be nonlocal in f .

Polarization gives the effective sesquilinear form $e^{\text{eff}}(g, f) = \langle A_{\omega,L}g, A_{\omega,L}f \rangle$. If this reduced form is densely defined and closed on $L^2(I_L)$, it determines a nonnegative self-adjoint boundary operator $S_{\omega,L}$, characterized by

$$e^{\text{eff}}(g, f) = \langle g, S_{\omega,L}f \rangle$$

for f in its operator domain and g in its form domain. When $A_{\omega,L}$ is taken as a closed factor, this is the operator $A_{\omega,L}^*A_{\omega,L}$. The boundary map Tr and the boundary operator $S_{\omega,L}$ thus have different jobs: one selects prescribed data, while the other describes the response after the bulk has equilibrated. Neither the trace nor the minimizing extension is assumed bounded in the ordinary L^2 norm merely because the notation resembles that of finite matrices.

The same mechanism in block form. The algebra can be seen completely in a finite-dimensional quadratic system. Separate the fixed boundary vector f from the free interior vector u , and let

$$\mathcal{E}[f, u] = \begin{pmatrix} f \\ u \end{pmatrix}^* \begin{pmatrix} H_{\text{bb}} & H_{\text{bi}} \\ H_{\text{ib}} & H_{\text{ii}} \end{pmatrix} \begin{pmatrix} f \\ u \end{pmatrix}, \quad H_{\text{ib}} = H_{\text{bi}}^*.$$

Assume the full block matrix is nonnegative and H_{ii} is positive definite. Here $\text{Tr}(f, u) = f$. Minimizing over u gives

$$u_{\min} = -H_{\text{ii}}^{-1}H_{\text{ib}}f, \quad S = H_{\text{bb}} - H_{\text{bi}}H_{\text{ii}}^{-1}H_{\text{ib}}.$$

Indeed, completing the square yields

$$\mathcal{E}[f, u] = \langle f, Sf \rangle + \|H_{\text{ii}}^{1/2}(u + H_{\text{ii}}^{-1}H_{\text{ib}}f)\|^2.$$

Thus $\mathcal{E}^{\text{eff}}[f] = \langle f, Sf \rangle$; the matrix S is the *Schur complement*. The subtracted term describes the energy reduction obtained by allowing the interior to relax. Positivity of S follows from positivity of

the *full* energy matrix. Positivity of the interior block alone would not suffice. Even if the original couplings are local or sparse, the inverse interior operator can couple distant boundary inputs. This is the algebraic reason that eliminating local bulk fields can produce a nonlocal boundary form. For differential operators or infinitely many channels, the same calculation requires a form-domain and convergence argument.

In a geometric extension this reduced response is often called a *Dirichlet-to-Neumann map*: prescribe the boundary value (Dirichlet data), solve the equilibrium equation, and read the outward normal flux (Neumann data). For a regular weighted energy $\int a(y)|\nabla U|^2 dx dy$, integration by parts in an equilibrium field identifies the effective energy with the boundary pairing against that flux, with the remaining boundary terms fixed by the domain. The trace itself only reads $U(x, 0)$; the flux response also depends on the solved field. Positive extension realizations of functions of the Laplace operator provide a rigorous setting for this mechanism [3].

What must match the arithmetic form. The qualifier *target* in (1.14) separates the variational construction from the arithmetic theorem still required. One must specify a bulk energy and its domains from independent data, solve or characterize the constrained minimum, and compute the resulting boundary form. That computation must reproduce $Q_{\omega, L}$ with its exact normalization and signs. For RH, the required central identity includes the gamma term, the pole terms and every active prime delay at arbitrarily large lengths. A successful match then supplies $Q_{0, L}[f] = \|A_{0, L}f\|^2$ and invokes (1.3). A positive bulk system by itself proves positivity of its own reduced form; identifying that form with the Weil form is the substantive research problem.

An explicit positive realization of the gamma kinetic term. We now derive the bulk mechanism for the kinetic component of the opening decomposition. The construction uses the positive partial fractions of the digamma function and does not assume the sign of the full Weil form. Recall $a_k = 2k + 1/2$ and

$$\begin{aligned} B(s) &= \sum_{k=0}^{\infty} \frac{2}{a_k} \frac{s}{a_k^2 + s}, \quad s \geq 0, \\ K[\tilde{f}] &= \frac{1}{2\pi} \int_{\mathbb{R}} B(\tau^2) |\hat{f}(\tau)|^2 d\tau. \end{aligned} \tag{1.15}$$

The partial-fraction calculation in the opening subsection identifies this series with the gamma multiplier. Equivalently, set

$$j(y) = \sum_{k=0}^{\infty} e^{-a_k y} = \frac{e^{-y/2}}{1 - e^{-2y}}, \quad y > 0.$$

Then

$$\begin{aligned} 2 \int_0^{\infty} e^{-ay} (1 - \cos(\tau y)) dy &= \frac{2}{a} \frac{\tau^2}{a^2 + \tau^2}, \\ B(\tau^2) &= 2 \int_0^{\infty} j(y) (1 - \cos(\tau y)) dy. \end{aligned}$$

Tonelli's theorem applies to these nonnegative integrands. Near zero, $j(y) \sim 1/(2y)$ and $1 - \cos(\tau y) = O(y^2)$; at infinity j decays exponentially. This verifies convergence of the integral.

The term *kinetic* describes the dependence on spatial variation. Here the response is nonlocal, with logarithmic growth at high frequency. It is not the ordinary local $\|f'\|^2$ energy. For example, writing $F = \tilde{f}$ and applying Plancherel,

$$K[F] = \int_0^{\infty} j(y) \|F - F(\cdot - y)\|_{L^2(\mathbb{R})}^2 dy.$$

The fields below realize this response by energies local in x before minimization.

For auxiliary fields $u_k \in H^1(\mathbb{R})$, define

$$\mathcal{E}^{\text{kin}}[\tilde{f}, (u_k)] = \sum_{k=0}^{\infty} \frac{2}{a_k} \int_{\mathbb{R}} \left(|\tilde{f} - u_k|^2 + a_k^{-2} |u'_k|^2 \right) dx. \quad (1.16)$$

Every term is nonnegative and local in x before elimination. In the general notation above, $\text{Tr}(\tilde{f}, (u_k)) = f$: the first component is fixed and the fields u_k are varied independently. Their weak equilibrium equations are

$$(1 - a_k^{-2} \partial_x^2) u_k = \tilde{f} \quad \text{on } \mathbb{R}.$$

Solving these equations gives the minimizing extension $\mathcal{P}f = (\tilde{f}, (u_k))$. Completion of the square at each Fourier frequency gives

$$\inf_{(u_k)} \mathcal{E}^{\text{kin}}[\tilde{f}, (u_k)] = K[\tilde{f}], \quad \hat{u}_k(\tau) = \frac{a_k^2}{a_k^2 + \tau^2} \hat{f}(\tau). \quad (1.17)$$

Here is the full one-channel calculation. At frequency τ , put $\lambda = \tau^2/a^2$, $h = \hat{f}(\tau)$ and $v = \hat{u}(\tau)$. Completing the square gives

$$|h - v|^2 + \lambda |v|^2 = (1 + \lambda) \left| v - \frac{h}{1 + \lambda} \right|^2 + \frac{\lambda}{1 + \lambda} |h|^2.$$

The unique minimizer and its real-space expression are

$$\hat{u}(\tau) = \frac{a^2}{a^2 + \tau^2} \hat{f}(\tau), \quad u(x) = \frac{a}{2} \int_{\mathbb{R}} e^{-a|x-y|} \tilde{f}(y) dy.$$

For every $f \in L^2(I_L)$ this solution belongs to $H^2(\mathbb{R})$, so it is an admissible H^1 field and solves the equilibrium equation. Its smoothing length is a^{-1} . Multiplying the minimum by $2/a$ and applying Plancherel gives

$$\inf_{u \in H^1(\mathbb{R})} \frac{2}{a} \int_{\mathbb{R}} \left(|\tilde{f} - u|^2 + a^{-2} |u'|^2 \right) dx = \frac{1}{2\pi} \int_{\mathbb{R}} \frac{2}{a} \frac{\tau^2}{a^2 + \tau^2} |\hat{f}(\tau)|^2 d\tau.$$

Every tower configuration has energy at least the sum of these channel minima. The family of channel minimizers attains that sum whenever it is finite. Monotone convergence of the nonnegative integrands gives the kinetic reduction above; the infimum and sum have not been interchanged without justification. Since $B(\tau^2) = \log(2 + |\tau|) + O(1)$, finite minimized energy is equivalent to $f \in \mathcal{D}_{\log, L}$. The fields extend over the whole line even when the input is supported in I_L , and their exterior tails are included. Truncating the fields at the input endpoints would change the reduced energy.

The corresponding bulk amplitude map is explicitly

$$\mathcal{C}(\tilde{f}, (u_k)) = \left(\sqrt{\frac{2}{a_k}} (\tilde{f} - u_k), \sqrt{\frac{2}{a_k^3}} u'_k \right)_{k \geq 0}.$$

Its target is the Hilbert direct sum of the displayed $L^2(\mathbb{R})$ components. Thus the factor produced by elimination is $A^{\text{kin}} f = \mathcal{C} \mathcal{P} f$, and $\|A^{\text{kin}} f\|^2 = K[\tilde{f}]$. The minimizing channel configuration has finite total energy precisely when the displayed kinetic integral is finite. This identifies the trace, equilibrium extension, amplitude map and reduced energy in one concrete instance of the general construction.

What this construction establishes for the program. The calculation realizes the arithmetic kinetic component as the minimum of an independently nonnegative local field energy:

$$K[\tilde{f}] = \inf_{(u_k)} \mathcal{E}^{\text{kin}}[\tilde{f}, (u_k)] = \|A^{\text{kin}} f\|^2 \geq 0.$$

This is the first constructive success of the proposed mechanism: one prescribed component is obtained exactly, with explicit field equations, an attained minimum and the required input domain. It also supplies a positive system whose couplings can be modified to seek the remaining arithmetic contributions. The masses and weights are chosen from the gamma identity; their realization does not show that supersymmetry selects them.

The full target is the boxed decomposition in the opening subsection. The full central Weil form additionally contains $w_0 \|f\|^2$, where $w_0 = \psi(1/4) - \log \pi < 0$, the pole terms, and the prime sum in (1.2). Explaining these contributions within a common positive bulk energy is the remaining completion problem. The kinetic construction does not by itself solve it.

Where the shift can enter a bulk description. There are two distinct uses of ω . In (1.14), it is a parameter of the energy and boundary problem; the direct RH target evaluates that family at $\omega = 0$. Alternatively, if the full generator factors are available for $0 \leq s \leq \omega$, the shift-energy identity becomes

$$\langle f, D_{\omega, L} f \rangle = 2 \int_0^\omega \|A_{s, L} V_{s, L} f\|_{\mathcal{K}_{s, L}}^2 ds. \quad (1.18)$$

This describes an accumulated positive energy on the strip $0 < s < \omega$, with initial data f and trajectory $g(s) = V_{s, L} f$. Its boundary observable is the contraction defect $D_{\omega, L}$; the central form is recovered from its initial rate by (1.9). This is a conditional identity: the full factors must first exist throughout the integration interval on the evolved inputs, with measurable and integrable squared amplitudes. The kinetic factor A^{kin} alone cannot be substituted for a full generator factor. An independent bulk explanation would also have to derive the factor and evolution from the specified system; writing the integral after those objects are known does not supply the missing arithmetic mechanism.

The coordinate x labels the original input, L limits its support, and a geometric coordinate y or channel index k labels additional degrees of freedom. The zeta shift ω is not automatically that geometric coordinate. Using the running shift s as an evolution coordinate in (1.18) is a specific interpretation with the displayed energy identity as its content. No AdS geometry, conformal field theory, or interacting supersymmetric field theory is assumed or established by these constructions.

References

- [1] M. Suzuki, *Weil's quadratic form via the screw function*, 2026, version 2, especially §1.1. [arXiv:2606.09096v2](#).
- [2] NIST Digital Library of Mathematical Functions, Chapter 5, especially §§5.4, 5.7, 5.9 and 5.11. [DLMF, Chapter 5](#).
- [3] M. Kwaśnicki and J. Mucha, *Extension technique for complete Bernstein functions of the Laplace operator*, Journal of Evolution Equations **18** (2018), 1341–1379. [doi:10.1007/s00028-018-0444-4](#); [arXiv:1707.02475](#), especially the variational formulation and its measure-coefficient extension.